5G Core Network Traffic Prediction Based on NWDAF Multi-Model Fusion

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***Abstract*—Large-scale device access in 5G may exceed core network resource constraints, affecting user demand. Analyzing and predicting core network traffic is vital to optimise network configuration, improve quality of service and reduce costs. However, accurately predicting network traffic while ensuring data privacy is a challenging task due to the diversity, complexity, and sensitivity of core network traffic data. Inspired by Network Data Analytics Function(NWDAF) and ensemble learning, we propose NWDAF-MMF, a multi-model fusion network traffic prediction framework based on NWDAF. In NWDAF-MMF, NWDAF obtains data from Network Functions(NFs), reducing the need for numerous data transmissions and storage. Additionally, multi-model fusion can process different types of data provided by NFs. Our method comprises three submodules: the data preprocessing module, the multi-model training and analysis module, and the multi-model fusion module. The data processing module is responsible for data aggregation, data cleaning, feature selection, and dataset division provided by NFs. The multi-model training and analysis module is responsible for training the base model and obtaining its prediction results. The multi-model fusion module is responsible for integrating, optimizing, and outputting the final prediction results. We evaluate the proposed method using real network datasets. The experimental results show improved performance in MAE, RSME, and MAPE by 12.32%, 6.40%, and 12.66% compared to the single model, respectively.**

***Index Terms*—5GC, Network Traffic Prediction, NWDAF, Multi-Model Fusion**

I. Introduction

As 5G technology becomes increasingly commercialized and popularized [1], the coverage and number of access devices in 5G networks continue to grow. 5G networks provide higher bandwidth and lower latency, enabling support for not only traditional mobile communication services but also various new application scenarios such as the internet of things, remote healthcare, vehicular networks, and smart cities [2]. However, in the face of a large number of device accesses, the network may fail to meet user demands due to its limited network resources [3]. Therefore, analyzing and forecasting network traffic in the 5G core network is crucial as it assists network operators in better planning network resources, optimizing network configuration, delivering high-quality services, and reducing operational costs.

In 5G core networks, processing extensive data and high-dimensional features is necessary to predict traffic due to the massive volume of data and diverse characteristics and types of network elements. Network traffic exhibits spatiotemporal correlations due to factors such as time and space, including day and night, weekdays and holidays, and urban and rural areas, thereby increasing the complexity of traffic prediction. Additionally, network traffic exhibits nonlinearity and uncertainty and is influenced by various factors such as weather, activities, and events, thereby making it challenging for traffic prediction models to capture the patterns of traffic fluctuations accurately. The prediction of user access traffic in 5G core networks is challenged by these real-world factors.

Many scholars have conducted extensive predictive research in the field of 5G networks. For example, Chakraborty et al. employed time series analysis of traffic logs in the 5G core network to predict potential threats that could result in system failures [4]. Ibrahimi and Serbouti utilized Auto-Regressive (AR), Moving Average (MA), and Exponential Smoothing techniques to forecast content popularity in 5G networks [5]. Nevertheless, these traditional methods demonstrate inferior predictive capabilities when dealing with data characterized by high dynamics, nonlinearity, and spatiotemporal features.

In contrast, machine learning methods are better equipped to handle complex, high-dimensional data and capture nonlinear relationships. Najm et al. [6] proposed a decision tree-based framework for predicting optimal nodes and managing congestion in 5G IoT scenarios. Similarly, Minovski et al. [7]

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employed diverse machine learning models to predict LTE cellular link performance. However, these machine learning approaches require a large volume of labeled training data and are impacted by parameter and model choices.

Due to the rapid advancement of computational power, an increasing number of researchers are employing deep learning methods for prediction, as these methods are better at capturing long-term dependencies and nonlinear changes in time-series data. For example, Paul et al. [8] utilized an RNN model to forecast the next day's traffic demand in various areas of Milan, Italy. Additionally, Wang et al. [9] performed spatiotemporal analysis on real-world cellular network traffic data and introduced a Time Series Generative Adversarial Network (TSGAN) based on temporal sequence similarity to predict spatiotemporal cellular network traffic. Furthermore, Alawe et al. extended predictions of traffic load variations in 5G core network resources using LSTM and DNN [10]. Nevertheless, the training and tuning of deep learning models are quite complex.

Network traffic data exhibits diversity and complexity, and a single model may not fully capture all data features and patterns. Moreover, network operators typically control network data and may involve user privacy and commercial secrets. Scholars often struggle to obtain sufficient data for research. We are surprised to find that 5G core network introduces NWDAF, which can solve the above problems well. The NWDAF model store can provide various models for model training to solve the limitation of a single model. The NWDAF model training data is provided by the NFs, which can be deployed near the NFs, solving the problem of difficulty in accessing the data. Therefore, we based on NWDAF for analysis of 5G core network traffic data and prediction. The contributions of this paper are summarized as follows.

- We propose a network traffic prediction method that uses multi-model fusion based on the property of NWDAF, which can store various models and obtain data from the NFs.
- Compared with a single model, our method combines the advantages of multiple models. Experimental results show that our method is better than the single model in MAE, RMAE, and MAPE metrics.
- Unlike plug-in AI methods, our method can locally directly analyze and predict the NFs data in the core network, reducing the privacy issues caused by data migration, storage, and processing.

The rest of this paper is structured as follows: Section II presents related work on correlation. Section III provides a detailed account of the model's construction method and derivation. Section IV outlines the experimental setting, results and analyses. Section V concludes the paper and discusses the future work prospects.

II. Related work

A. NWDAF

The 3rd Generation Partnership Project (3GPP) introduced the NWDAF in Release 15 as an integral component of the 5G core network architecture. NWDAF relies on artificial intelligence (AI) and machine learning (ML) techniques to offer specialized network data analysis services [11]. The architecture for NWDAF's data collection and analysis exposure is depicted in Figure 1.

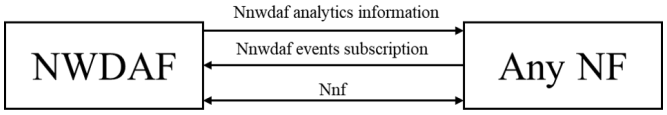

Fig. 1: NWDAF data collection and analysis exposure architecture.

The NF can request a data delivery subscription from NWDAF via the Nnwdaf interface. NWDAF can furnish network analysis data to the NF. NWDAF can retrieve data from the NF using the Nnf interface.

NWDAF, functioning as an AI NF within the core network, collects data from diverse NFs and OAM, which serves as the foundational dataset for network analytics, as shown in Figure2.

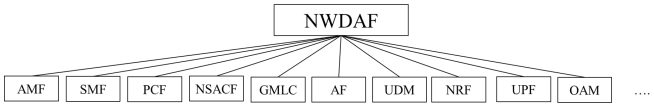

Fig. 2: Data sources that NWDAF can collect.

NWDAF contains model training logic functions(MTLF) and analysis logic functions(AnLf). MTLF is responsible for training ML models and exposing them to AnLF. AnLF is responsible for reasoning, reporting, and exposing the analysis service [12]. The NWDAF training ML model architecture is shown in Figure 3.

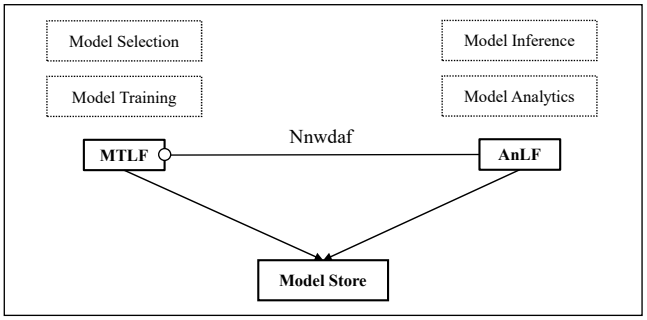

Fig. 3: NWDAF training ML model architecture.

MTLF can select models from Model Store during model training, which provides the basis for the multi-model fusion proposed in this paper.

Presently, NWDAF's achievable use case scenarios encompass slice load, observed service experience, NF load analytics, network performance analytics, UE-related analytics, user data congestion analytics, QoS sustainability analytics, dispersion analytics, WLAN performance analytics, session management

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congestion control experience analytics, redundant transmission experience related analytics, DN performance analytics, PFD determination analytics, location accuracy analytics, End-to-end data volume transfer time analytics, relative proximity analytics, PDU session traffic analytics, movement behavior analytics. [13].

B. Traffic Prediction Based on Multi-model Fusion

Traffic prediction encompasses the projection of future traffic levels based on historical data and relevant features, finding wide-ranging applications in industries such as finance, retail, transportation, and telecommunications [14].

Network traffic data typically exhibits time series characteristics. Common models include Autoregressive Integrated Moving Average(ARIMA), Seasonal Autoregressive Integrated Moving Average(SARIMA) [15], and Prophet [16] among traditional models; Decision Trees [17], Random Forests [18], Support Vector Machines(SVM) [19] among machine learning models; Long Short-Term Memory(LSTM) [20] and Gated Recurrent Units(GRU) [21] as deep learning models. Each model possesses unique advantages and drawbacks. Traditional models are straightforward but have strict data distribution assumptions, while machine learning models can handle complex relationships but may necessitate substantial feature engineering. Deep learning models excel in managing complex data but require considerable data and computational resources.

The evolution of machine learning and data mining has advanced the maturity of model fusion techniques, providing methods such as Bagging [22], Boosting [23], Stacking [24], and weighted averaging. This paper exploits the attributes of multiple models, comprehensively evaluating their respective strengths and weaknesses with the support of NWDAF, thus enhancing the accuracy, robustness, and generalization ability of traffic prediction and better addressing complex forecasting tasks.

III. Methodology

A. Overview

To make the core network traffic prediction more comprehensible, as shown in Figure 4, we propose the network traffic prediction method NWDAF-MMF. It consists of three modules, which are the data preprocessing module (1), the multi-model training and analysis module (2), and the multi-model fusion module (3).

As shown in Figure 4, the NWDAF in the core network can interact with various NFs such as AMF, SMF, and UDM. Our research objective is to predict the future core network traffic using various UE data (e.g., mobile phones, cars, smart wearable devices) collected by the gNB. NFs provide the data for model training, and the NWDAF in the core network aggregates and preprocesses the data selects appropriate models for training, and provides analysis results. Unlike previous research methods, we take advantage of the fact that NWDAF can store models, select multiple models for training, and integrate the prediction results of different models through ensemble learning to improve the model's generalization ability. The problem is defined as follows.

Given the signaling data for a particular NF as X, the data X is preprocessed(X_{clean}) for feature selection($X_{selected}$), and then the dataset is divided into a training(X_{train}) and test(X_{test}) set. Our goal is to request multiple models(M_i) from NF to NWDAF to be trained separately and output their respective prediction results(Y_i), and finally output the final prediction results(Y) through the multi-model fusion module. The specific derivation of NWDAF-MMF will be explained in the next sections.

B. Data Preprocessing Module

Data preprocessing involves the manipulation of raw data before machine learning or data analysis. This process aims to enhance data quality, model performance, accuracy, and generalization.

In our framework, the signalling data collected by the gNB from the UE contains various feature information, but some feature information does not affect model training. According to the definition of 3GPP, NWDAF obtains statistical information from NFs, and different types of data are needed for model training in different NWDAF scenarios, so the signalling data cannot be used directly by NWDAF. The NFs need to perform statistical information on the signalling data obtained from the gNB, e.g., the time when the UEs accessed or left the gNB is the critical feature information in this study. Due to factors such as technical problems or environmental conditions (including weak signals, network congestion, and data transmission errors) may lead to incomplete signalling information in the data collected by the gNB. The NF must delete the signalling data with missing key features when counting the information, as shown in Equation 1.

$$X_{new} = X_{old} - \{x_i | x_i = NAN\}, \tag{1}$$

where x_i represents data with incomplete information, X_{old} represents the original dataset, and X_{new} represents the dataset obtained after removing the x_i.

After removing the x_i data, we merged multiple data sources into one dataset(X_{clean}). Subsequently, the time step(TS) is selected as the feature to obtain a new dataset($X_{selected}$) containing TS and user traffic(UT). To mitigate anomalies in the UT data collection process attributed to technical and environmental factors related to TS, this paper employs Z-score outlier detection for anomaly processing, as shown in Equation 2.

$$z = \frac{UT_t - \frac{1}{n}\sum_{i=1}^{n} UT_i}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(UT_i - \overline{UT})^2}}, \tag{2}$$

where UT_t represents the user traffic value of the dataset at moment t, and $\overline{UT}$ denotes the mean user traffic value of the dataset. Equation 3 describes the handling of outliers.

$$UT_t = \begin{cases} \overline{UT} & z > c, \\ UT_t & Other, \end{cases} \tag{3}$$

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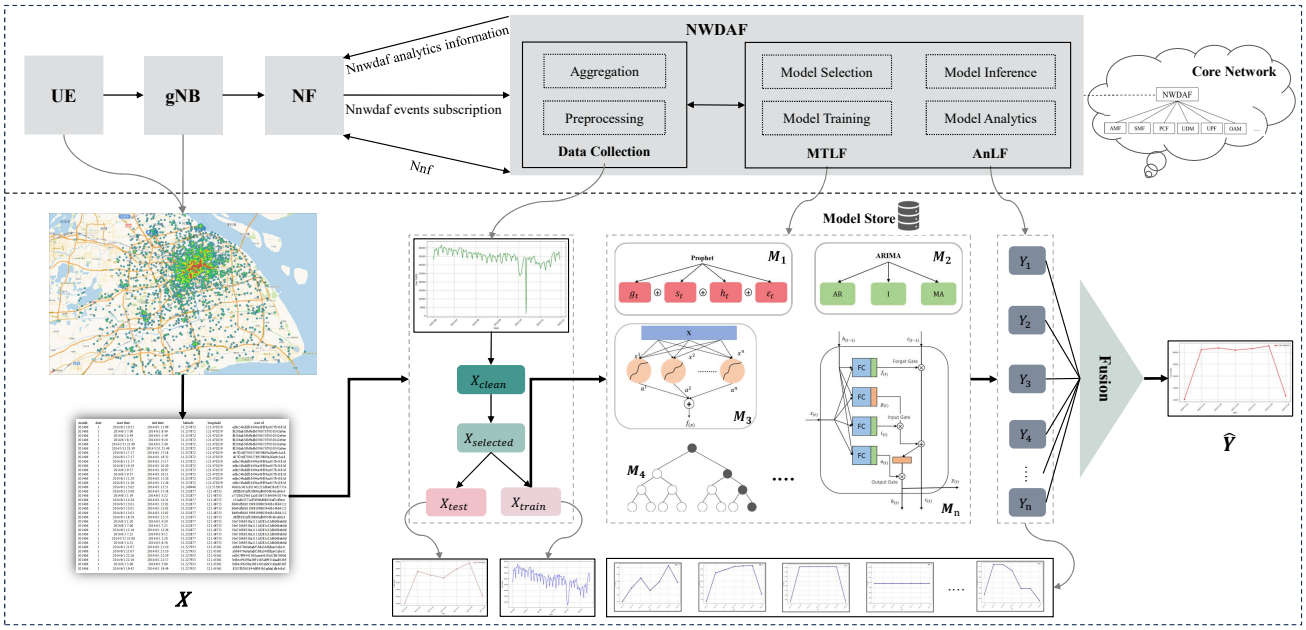

Fig. 4: NWDAF-MMF framework.

where c is the outlier threshold.

Data normalization standardizes the range of values, mitigates the influence of varying magnitudes among different features, enhances algorithm stability, and accelerates convergence. Min-max normalization is applied specifically to normalize the data related to the variable UT, given its significant magnitude within the time series dataset, as shown in Equation 4.

$$UT_{norm} = \frac{UT - UT_{\min}}{UT_{\max} - UT_{\min}}. \tag{4}$$

After the model has predicted the results, the predictions need to be back-normalized in order to get the true prediction and back-normalized using the min-max normalization method, as shown in Equation 5,

$$UT = (UT_{norm} \times (UT_{max} - UT_{min})) + UT_{min}. \tag{5}$$

Lastly, the dataset is divided as shown in Equation 6,

$$N = T + R, \tag{6}$$

where N represents the total number of samples, R represents the training set size, and T represents the test set size.

C. Multi-Model Training and Analysis Module

Section II-A defines NF as a subscriber to data analysis functions from NWDAF via *NWDAF event subscription*, and MTLF is responsible for model training. Different subscription requirements (classification, anomaly detection, prediction.) of NF require that NWDAF integrate multiple models to satisfy NF's needs. This paper uses this feature of NWDAF to subscribe to a single NF requirement in NWDAF, train multiple models of the same type simultaneously in MTLF, and finally fuse the prediction results of multiple models to improve the generalization ability of accuracy and reduce the selection dependency on algorithms.

This module selects common time series prediction models for training, containing traditional machine learning and deep learning models. The MMTA's model set can be defined as Equation 7,

$$MMTA_{mod} = \{M_1, M_2, M_3, M_4, M_5, \cdots, M_n\}. \tag{7}$$

Each model in MMTA requires parameter tuning and optimization, and then the model is trained for prediction. Its training process is shown in Algorithm 1.

The algorithm commences by receiving the training and test datasets, along with the corresponding model parameters. Within the MMTA framework, each base model is initialized, fitted, and compiled. This process yields the predicted values, which are then output for subsequent evaluation.

$$MMTA_{pre} = \{Y_1, Y_2, Y_3, Y_4, Y_5, \cdots, Y_n\}. \tag{8}$$

D. Multi-Model Fusion Module

During the model training process, a single model may face challenges in generalization when dealing with a variety of problems and datasets. Nevertheless, by employing model fusion, the unique strengths of individual models can be combined to attain the optimal solution. The outcomes vary based on the specific model fusion method, process, or result fusion selected to tackle a particular problem [24]. In our proposed methodology, the complexities of the base model types and their low correlation present difficulties in standardizing inputs, parameter selection, and training phases. Consequently, we fuse the results and assign prediction weights to each base model using the random search method.

In this subsection, the optimal weight combinations of the base model are determined through the model fusion method to

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Algorithm 1 MMTA for UT Prediction

Input:
$\{M_1, M_2, M_3, M_4, M_5, \cdots, M_n\}$: $MMTA_{mod}$
X: Input time series data
Y: Corresponding target values
X_{test}: Test data for prediction
// M_{i_p} is the parameter of the model in MMTA
{growth, seasonality, holidays, interval_width, $\cdots$}: M_{1_p}
$\{p, d, q\}$: M_{2_p}
{Max depth, Min samples split, Min samples leaf}: M_{3_p}
{kernel, C, epsilon, gamma}: M_{4_p}
{activation, input_shape, units, batch_size, $\cdots$}: M_{5_p}, M_{6_p}
Output:
$\{Y_1, Y_2, Y_3, Y_4, Y_5, \cdots, Y_n\}$: $MMTA_{pre}$

```
for i in n:
    Model_{M_i} = M_i(parameters)
    if M_i != LSTM or GRU:
        Model_{M_i} · fit(X, Y)
    else:
        Model_{M_i} · compile()
        Model_{M_i} · fit(X, Y)
    Y_i = Model_{M_i} · predict(X_test)
return Y_i
```

derive the final prediction results, as facilitated by Algorithm 2.

Algorithm 2 Optimal weights based on random search

Input:
$\{Y_1, Y_2, Y_3, Y_4, Y_5, \cdots, Y_n\}$: Base model prediction results
$\frac{1}{n}\sum_{i=1}^{n} Y_i$: Optimising target results
$num_{iterations}$: Number of searches
Output:
$\{\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \cdots, \alpha_n\}$: Best weights

```
best_performance ← ∞
best_weights ← ∅
for i in num_iterations:
    weights ~ Dirichlet(α_1, α_2, ···, α_n)
    weight_average ← Σ_{i=1}^{n} (Y_i × weight_i)
    RMSE ← sqrt(1/n Σ_{i=1}^{n} (Opt_{results_i} − weight_{average_i})^2)
    if RMSE<best_performance:
        best_performance ← RMSE
        best_weights ← weights
return best_weights
```

Algorithm 2 uses the predicted results and mean values of the MMTA as inputs and the RMSE as an optimization function by random search method to finally obtain the best weight combination.

$$Best_{wei} = \{\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \cdots, \alpha_n\}. \tag{9}$$

Therefore, the final prediction result($\hat{Y}_i$) is shown in Equation 10,

$$\hat{Y}_i = \sum_{i=1}^{n} \alpha_i \times Y_i. \tag{10}$$

IV. Experiments

In this section, we present comparative experiments of various models to assess the effectiveness of the developed NWDAF-MMF. We provide the dataset and commonly used evaluation metrics, followed by an outline of the model parameters used in the comparative experiments. Lastly, we present the experimental results of both the base model and the fusion model, followed by their analysis.

A. Dataset

To objectively assess the performance of the method, we conducted experiments on the authentic Telecom Shanghai Dataset. The dataset, generously provided by Shanghai Telecom, encompasses over 7.2 million records of internet access attempts made through 3,233 base stations and 9,481 mobile phones over a period of six months. The Telecom dataset encompasses six distinct parameters, including Month, Data, Start Time, End Time, Base Station Location, and Mobile Phone ID [25]. Researchers have leveraged this dataset to explore solutions in mobile edge computing [26]–[28].

Before model training, we need to process the dataset through the data preprocessing module. Firstly, all the signalling data collected by gNB are nulled to remove the data that lacks start time and end time. Then, all the data are classified in time steps of days, and the daily user traffic is recorded to form a dataset containing Date, Date type, and User Traffic. Finally, User Traffic is normalised and used as data for model training.

Based on the actual data, we extract the date and date type as feature columns to train the model. Next, we evaluate the relationship between features and user traffic using the Pearson correlation coefficient, as shown in Figure 5. A correlation coefficient value between 0.4 and 0.6 is generally considered to be moderately correlated, while 0.6-0.8 is strongly correlated. It is evident from the figure that the correlation coefficient between the date and user traffic is 0.66, indicating a strong correlation. Similarly, the correlation coefficient between date type and user flow is 0.42, also indicating a strong correlation. Therefore, it is reasonable to choose the date and date type as features to train the model in the experiment.

B. Evaluation metrics

This paper employs three commonly used evaluation metrics in traffic forecasting to assess the model: mean absolute error (MAE), root mean square error (RMSE), and mean absolute percentage error (MAPE).

- MAE represents the average absolute difference between the predicted and actual values, thus reflecting the overall

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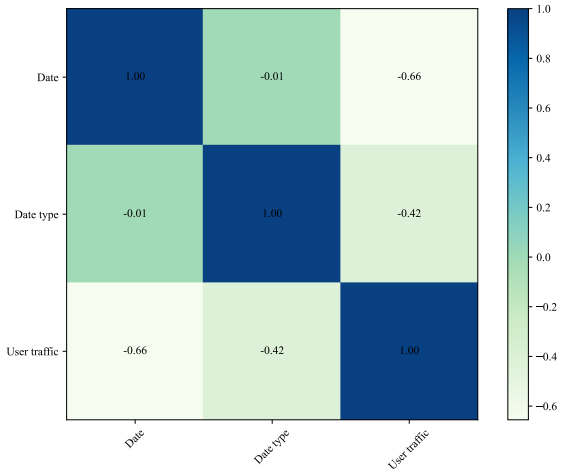

Fig. 5: Correlation Matrix Heatmap: Visualise the relationship among Data, Data type, and User Traffic.

prediction accuracy of the model. The calculation equation is as follows:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|, \tag{11}$$

where n is the number of samples, y_i is the actual value, and $\hat{y}_i$ is the corresponding predicted value. A smaller MAE value indicates a more accurate prediction by the model.

- RMSE is the root mean square of the difference between the predicted and actual values. The calculation equation is as follows:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}. \tag{12}$$

RMSE reflects the stability and accuracy of the model prediction. Similarly, the smaller the value of RMSE, the more accurate the model prediction.

- MAPE is the mean percentage difference between predicted and actual values, reflecting the relative accuracy of the model at different scales. The equation is as follows:

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{|y_i - \hat{y}_i|}{y_i}\right) \times 100\%. \tag{13}$$

C. Parameter Setting

In our framework, the base model selection of time series forecasting models includes traditional models (Prophet and ARIMA), machine learning models (DT and SVR), and deep learning models (LSTM and GRU) to ensure that different types of models are covered. To comply with the reality of this study, the fusion models, such as bagging, boosting, and weighted averaging, were selected to fuse the results of the base models, where random search determined the parameters of weighted averaging.

The parameter settings impact the experimental results of the model and ensure the reliability and repeatability of the experiment. Table I lists the model parameters for Prophet, ARIMA, DT, SVR, LSTM, and GRU, along with their corresponding value settings. Similarly, Table II includes the model parameters for Bagging, Boosting, and Random Search, along with their respective value settings. Due to the large number of model parameters, only the key parameters are included in the tables, while the parameters set to default values or those that do not affect the model results are excluded.

TABLE I: The settings of base model parameter

Model	Parameter	Meaning	Setting
Prophet	growth	Trend	linear
	interval_width	Confidence interval	0.95
	weekly_seasonality	Weekly seasonality	23
	holidays_prior_scale	Holidays, seasonal change points prior scalar factors	15
	changepoint_prior_scale		25
	seasonality_prior_scale		15
ARIMA	p	Autoregressive order	4
	d	Differencing order	1
	q	Moving average order	10
DT	max_depth	Maximum depth	8
	min_samples_split	Minimum samples for splitting and leaf node	2
	min_samples_leaf		1
SVR	kernel	Kernel function	rbf
	C	Penalty coefficient	1
	epsilon	Slack variable	100
LSTM GRU	activation	Activation function	relu
	input_shape	Input layer shape	(1, 2)
	epochs	Training rounds	200
	batch_size	Batch size	128

TABLE II: The settings of fusion model parameter

Model	Parameter	Meaning	Setting
Bagging	n_estimators	Number of basic models	6
	num_bagging_runs	Number of Bagging	100
	model	Training Models	Ridge()
Boosting	n_estimators	Number of basic models	6
	learning_rate	Learning rate	1
	max_depth	Maximum depth	3
Our Method	num_iterations	Number of iterations	100
	weights	Base model weights	Dirichlet

D. Experimental Results and Analysis

Figure 6 depicts the observed user traffic trends. The figure indicates that user traffic exhibits a consistent overall trend, with fluctuations occurring on a weekly (7-day) basis. Moreover, user traffic experiences significant changes during holidays, although holidays are infrequent, and the traffic promptly reverts to its previous level post-holidays. To assess

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the impact of various models in our experiments, we utilize the prediction results from the final 7 days (2014-11-23 to 2014-11-29) for verification purposes.

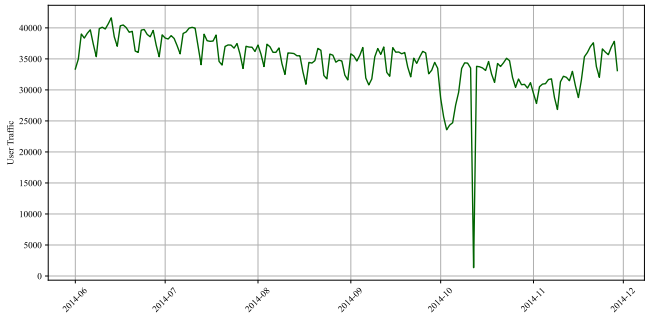
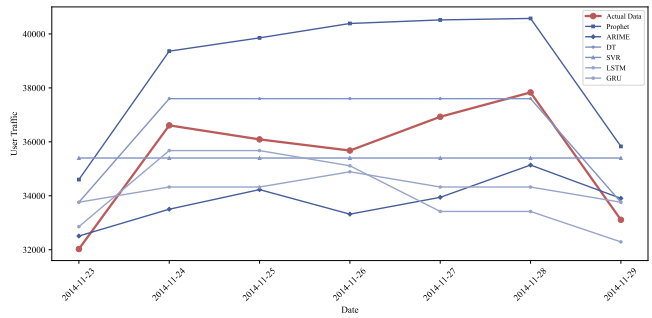
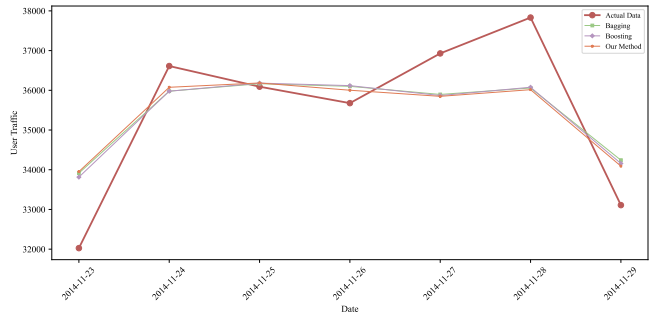


Fig. 6: Actual user traffic data



(a) Base model prediction results



(b) Fusion model prediction results

Fig. 7: Actual user traffic data and model prediction results

Figure 7a illustrates the prediction results obtained from the six base models. Analysis of the figure reveals that the prediction results of the Prophet model exhibit an overall trend higher than the actual data. While the predictions align well with the real data trend during the initial and final four days, discrepancies arise in the middle three days. In contrast to the Prophet model, the predictions made by the ARIMA model demonstrate strong alignment with the actual data during the middle four days. However, there is a variation in the second day's data, resulting in an overall trend lower than the actual data. The prediction results of the DT model display a general trend similar to the actual data, with complete consistency and no changes observed during the middle four days. On the other hand, the SVR model predicts a linear trend, indicating the model's failure to capture the underlying data pattern effectively. The prediction result of the LSTM model exhibits a sudden deviation from the real data on the fourth day. Lastly, the prediction results of the GRU model align with the trend direction of the actual data during the initial four days, but afterward, a downward trend is observed.

Figure 7b presents the prediction results obtained from the three fusion models. Analysis of the figure reveals that the three models exhibit a similar overall trend and demonstrate closer alignment with the actual data compared to the base model. In comparison to Bagging and Boosting, our method achieves better alignment with the actual data on the 2nd, 3rd, 4th, and 7th day.

For a more precise comparison of the effects of different models, the evaluation metric results are presented in Figure 8. The figure demonstrates that the fusion model yields significantly superior results compared to the base model across the MAE, RSME, and MAPE metrics. Table III provides more detailed data, highlighting that our proposed method attains optimum values in terms of MAE and MAPE. Our method exhibits improvements of 12.32%, 6.40%, and 12.66% in terms of MAE, RSME, and MAPE, respectively, compared to the optimal result achieved by the DT model in the base model.

TABLE III: Evaluation metrics results

Models Classification		Model	MAE	RSME	MAPE
Base Model	Trad	Prophet	3346.41	3265.26	9.20%
		ARIME	2040.53	2258.99	5.62%
	ML	DT	1101.43	1247.44	3.16%
		SVR	1686.21	1960.38	4.86%
	DL	LSTM	1905.19	2120.45	5.29%
		GRU	1640.75	2220.56	4.50%
Fusion Model	EL	Bagging	993.96	1173.89	2.84%
		Boosting	973.93	**1140.63**	2.78%
		Our Method	**965.72**	1167.64	**2.76%**

V. Conclusion

This paper studies the problem of network traffic prediction. Most existing methods concentrate on a single model and plug-in methods for predicting network data, which are limited in their ability to generalize models to address data complexity and overlook privacy issues arising from multiple data transmissions and storage. We innovatively propose NWDAF-MMF. Our method consists of three parts: the data preprocessing module, the multi-model training and analysis module, and the multi-model fusion module. Through the joint work of the three modules, our method can effectively manage various types of data in NF and reduce data transmissions. At the same time, the experiments are established on the Shanghai Telecom dataset, and the experimental indicated that

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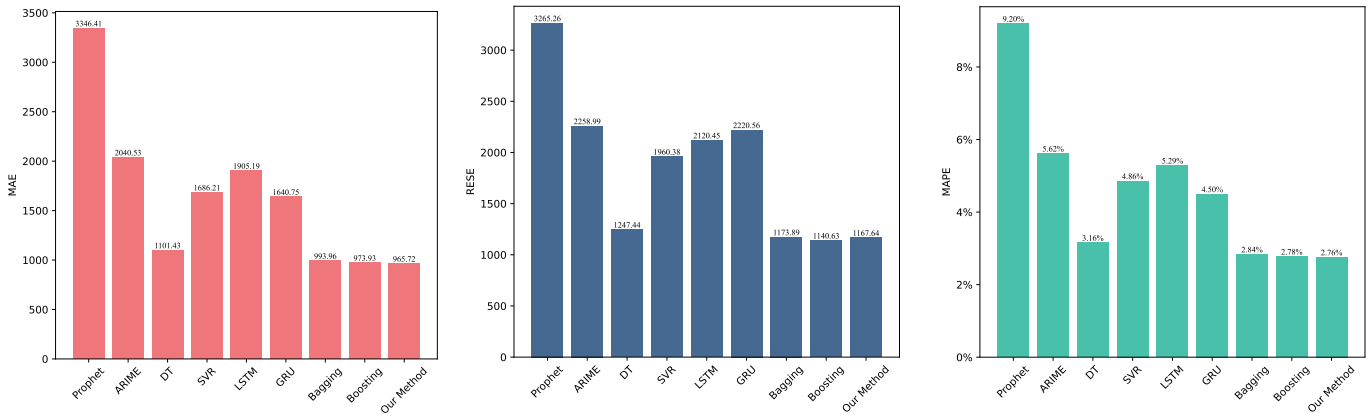

(a) MAE (b) RSME (c) MAPE

Fig. 8: Evaluation metrics results

our method is ahead of a single model in multiple metrics. This also proves the effectiveness of our method.

In the future, we will further improve the method to match the real data and reduce the errors and accommodate diverse data types from NFs. We will also explore training with federated learning models to avoid privacy issues caused by data transmission.

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